

# @modelcontextprotocol/server-filesystem (official)

✓ SHIP-READY — all MUST checks pass, no security findings, no behavioural drift

Server under test

npx -y @modelcontextprotocol/server-filesystem /tmp/mcp-proof-sandbox

Generated

2026-08-20 15:21 UTC · mcp-proof v0.2.0

Protocol

negotiated 2025-11-25 · latest spec is 2026-07-28 — migration recommended

Run fingerprint

sha256:b01888e9b1e702b033048412a30239a105107b08e534bdbb50d40ab8a51a7601

CONFORMANCE

11/11

MUST checks passed · 4/4  
SHOULD

SECURITY

1

finding · 0 blocking · 1 advisory

BEHAVIOUR REGRESSION

34/34

replays clean · gate PASS

Protocol conformance

| CHECK   | LEVEL  | RESULT | DETAILS                                                                                                                                                                                                                                                              |
|---------|--------|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| LIFE-01 | MUST   | ✓ PASS | <b>initialize returns protocolVersion, capabilities and serverInfo</b><br>protocolVersion, capabilities and serverInfo all present                                                                                                                                   |
| LIFE-02 | SHOULD | ! WARN | <b>server negotiates the latest spec revision (2026-07-28)</b><br>requested 2026-07-28, negotiated 2025-11-25 — server not yet migrated to the 2026-07-28 spec revision<br><b>Fix:</b> Upgrade the server SDK to a release that speaks the 2026-07-28 spec revision. |
| LIFE-03 | MUST   | ✓ PASS | <b>server answers tools/list after the initialized notification</b><br>tools/list returned 14 tool(s) after initialized                                                                                                                                              |
| LIST-01 | MUST   | ✓ PASS | <b>tools/list pagination terminates (no cursor loop)</b><br>single page, no pagination cursor                                                                                                                                                                        |
| RPC-01  | MUST   | ✓ PASS | <b>unknown method gets a JSON-RPC error response</b><br>unknown method rejected with error code -32601                                                                                                                                                               |
| RPC-02  | SHOULD | ✓ PASS | <b>unknown method error code is -32601 (method not found)</b><br>expected -32601, got -32601                                                                                                                                                                         |

| CHECK   | LEVEL  | RESULT | DETAILS                                                                                                                                                                                          |
|---------|--------|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| RPC-03  | MUST   | ✓ PASS | <b>malformed tools/call params are rejected with an error</b><br>malformed params rejected with error code -32603                                                                                |
| T00L-01 | MUST   | ✓ PASS | <b>every tool has a non-empty name and an inputSchema</b><br>all 14 tools have a name and an inputSchema                                                                                         |
| T00L-02 | SHOULD | ✓ PASS | <b>every tool has a non-empty description</b><br>all 14 tools carry a description                                                                                                                |
| T00L-03 | MUST   | ✓ PASS | <b>every tool inputSchema compiles as JSON Schema</b><br>all 14 inputSchemas compile as JSON Schema draft 2020-12                                                                                |
| T00L-04 | MUST   | ✓ PASS | <b>calling a nonexistent tool is rejected</b><br>rejected as tool result with isError=true                                                                                                       |
| T00L-05 | MUST   | ✓ PASS | <b>a call missing required arguments is rejected</b><br>read_file with empty args rejected via isError=true                                                                                      |
| T00L-06 | MUST   | ✓ PASS | <b>declared outputSchemas compile and structured results match them</b><br>all 14 outputSchema(s) compile; dynamic call to read_file did not yield a normal result, structured output unverified |
| HYG-01  | MUST   | ✓ PASS | <b>stdout carries only JSON-RPC messages</b><br>no non-JSON-RPC stdout lines observed                                                                                                            |
| CAP-01  | SHOULD | ✓ PASS | <b>declared capabilities match served features (tools)</b><br>capabilities.tools declared and tools/list served                                                                                  |

## Security & hygiene

| CHECK  | DOMAIN       | RESULT | DETAILS                                                                                                   |
|--------|--------------|--------|-----------------------------------------------------------------------------------------------------------|
| SEC-01 | MCP-INPUT-01 | ✓ PASS | <b>no prompt-injection patterns in tool descriptions</b><br>0 matches across 14 tools                     |
| SEC-02 | MCP-INPUT-01 | ✓ PASS | <b>no invisible or bidi control characters in tool metadata</b><br>0 invisible characters across 14 tools |
| SEC-03 | MCP-LOG-02   | ✓ PASS | <b>no secret-looking strings in tool metadata</b><br>0 secret-like strings across 14 tools                |

| CHECK  | DOMAIN                  | RESULT            | DETAILS                                                                                                                                                                                                                                                                                                                                                                                                                      |
|--------|-------------------------|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SEC-04 | MCP-INPUT-02            | <div>! WARN</div> | <b>injection-surface string params carry constraints</b><br>unconstrained injection-surface params: read_file.path, read_text_file.path, read_media_file.path, write_file.path, edit_file.path, create_directory.path, list_directory.path, list_directory_with_sizes.path, directory_tree.path, search_files.path, get_file_info.path<br><b>Fix:</b> Add enum, pattern or maxLength to path/url/command-like string params. |
| SEC-05 | MCP-INPUT-01            | <div>✓ PASS</div> | <b>tool descriptions stay under 2000 chars</b><br>longest description 457 chars across 14 tools                                                                                                                                                                                                                                                                                                                              |
| SEC-06 | MCP-EXEC-01,MCP-EXEC-02 | <div>✓ PASS</div> | <b>no tool advertises unconstrained arbitrary execution</b><br>0 exec-style tools with free-form params across 14 tools                                                                                                                                                                                                                                                                                                      |

MSSS compliance

**L1: 2/2 auto-assessable controls met · 4 require manual review.** Mapped against MSSS v0.1 (control-level mapping v2.0 (2026-01-20)): 3 of 24 controls are auto-assessable from this audit's deterministic checks; the remaining 21 need deployment, code or process evidence and are marked manual review — never assessed by this tool.

| CONTROL      | DOMAIN           | LEVEL | RESULT                     | TITLE                                                                                                                                                                                                                                         |
|--------------|------------------|-------|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MCP-EXEC-01  | Execution        | L 1   | <div>– manual review</div> | <b>Prohibition of Shell Execution</b><br>SEC-06 flags advertised arbitrary-exec tools, but proving the absence of shell invocation requires source review.                                                                                    |
| MCP-FS-01    | Filesystem       | L 1   | <div>– manual review</div> | <b>Path Allowlisting and Canonical Resolution</b>                                                                                                                                                                                             |
| MCP-FS-02    | Filesystem       | L 1   | <div>– manual review</div> | <b>Symlink Resolution Validation</b>                                                                                                                                                                                                          |
| MCP-NET-01   | Network          | L 1   | <div>– manual review</div> | <b>URL Validation and SSRF Mitigation</b>                                                                                                                                                                                                     |
| MCP-INPUT-01 | Input Validation | L 1   | <div>✓ met</div>           | <b>JSON Schema Validation</b><br><b>Evidence checks:</b> TOOL-01, TOOL-03, RPC-03, TOOL-05, SEC-01, SEC-02, SEC-05<br>Auto-assessed from advertised schemas (present and valid), live rejection probes, and the tool-metadata poisoning scan. |
| MCP-LOG-02   | Logging          | L 1   | <div>✓ met</div>           | <b>Secret Redaction in Logs</b><br><b>Evidence checks:</b> SEC-03<br>Auto-assessed over advertised tool metadata; server log output is not inspected.                                                                                         |

| CONTROL       | DOMAIN           | LEVEL | RESULT          | TITLE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|---------------|------------------|-------|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MCP-SUPPLY-02 | Supply Chain     | L2    | – manual review | <b>Trusted Package Sources</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| MCP-INPUT-02  | Input Validation | L2    | ✓ met           | <b>Input Bounds Enforcement</b><br>Evidence checks: SEC-04<br>SEC-04: unconstrained injection-surface<br>params: read_file.path, read_text_file.path,<br>read_media_file.path, write_file.path,<br>edit_file.path, create_directory.path,<br>list_directory.path,<br>list_directory_with_sizes.path,<br>directory_tree.path, search_files.path,<br>get_file_info.path<br>Auto-assessed from advertised schema<br>constraints (enum/pattern/maxLength); runtime<br>payload limits are not probed. |
| MCP-INPUT-03  | Input Validation | L2    | – manual review | <b>Timeout Enforcement</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| MCP-NET-03    | Network          | L2    | – manual review | <b>TLS Enforcement</b><br>MSSS marks this N/A for stdio-only<br>deployments with no network access.                                                                                                                                                                                                                                                                                                                                                                                              |
| MCP-EXEC-02   | Execution        | L2    | – manual review | <b>Command Allowlisting</b><br>See the SEC-06 advisory for advertised<br>command tools; allowlist verification requires<br>source review.                                                                                                                                                                                                                                                                                                                                                        |
| MCP-EXEC-03   | Execution        | L2    | – manual review | <b>Argument Separator Usage</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| MCP-AUTHZ-01  | Authorization    | L3    | – manual review | <b>OAuth Token Delegation</b><br>MSSS marks this N/A for stdio transport with<br>OS-level user isolation.                                                                                                                                                                                                                                                                                                                                                                                        |
| MCP-AUTHZ-02  | Authorization    | L3    | – manual review | <b>Per-Tool Scope Definition</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| MCP-AUTHZ-03  | Authorization    | L3    | – manual review | <b>Least Privilege Tool Design</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| MCP-AUTHZ-04  | Authorization    | L3    | – manual review | <b>Resource-Based Access Control</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| MCP-LOG-01    | Logging          | L3    | – manual review | <b>Comprehensive Audit Logging</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| MCP-DEPLOY-01 | Deployment       | L3    | – manual review | <b>Container Hardening</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| MCP-FS-03     | Filesystem       | L4    | – manual review | <b>Filesystem Sandboxing</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|               |                  |       |                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

| CONTROL       | DOMAIN       | LEVEL | RESULT          | TITLE                                                                                                     |
|---------------|--------------|-------|-----------------|-----------------------------------------------------------------------------------------------------------|
| MCP-SUPPLY-01 | Supply Chain | L4    | – manual review | <b>Package Integrity Verification</b>                                                                     |
| MCP-DEPLOY-03 | Deployment   | L4    | – manual review | <b>Resource Limits and Rate Limiting</b>                                                                  |
| MCP-NET-02    | Network      | L4    | – manual review | <b>Egress Traffic Filtering</b>                                                                           |
| MCP-DEPLOY-02 | Deployment   | L4    | – manual review | <b>System Call Filtering (seccomp/AppArmor)</b>                                                           |
| MCP-DEPLOY-04 | Deployment   | L4    | – manual review | <b>Runtime Integrity Monitoring</b><br>Future control in MSSS v0.1 — implementation details TBD upstream. |

Control taxonomy from the MCP Server Security Standard (MSSS), CC BY-SA 4.0, mcp-security-standard.org.

## Behaviour regression

| FIXTURE                        | TOOL            | VERDICT | DETAIL |
|--------------------------------|-----------------|---------|--------|
| read_file__e8cf9eda.json       | read_file       | OK      |        |
| read_file__0c5f5c79.json       | read_file       | OK      |        |
| read_file__fe0dcdc6.json       | read_file       | OK      |        |
| read_file__518341a7.json       | read_file       | OK      |        |
| read_text_file__e8cf9eda.json  | read_text_file  | OK      |        |
| read_text_file__0c5f5c79.json  | read_text_file  | OK      |        |
| read_text_file__fe0dcdc6.json  | read_text_file  | OK      |        |
| read_text_file__518341a7.json  | read_text_file  | OK      |        |
| read_media_file__e8cf9eda.json | read_media_file | OK      |        |
| read_media_file__0c5f5c79.json | read_media_file | OK      |        |
| read_media_file__fe0dcdc6.json | read_media_file | OK      |        |
| read_media_file__518341a7.json | read_media_file | OK      |        |
|                                |                 |         |        |

| FIXTURE                                   | TOOL                      | VERDICT | DETAIL |
|-------------------------------------------|---------------------------|---------|--------|
| read_multiple_files__dfea4edf.json        | read_multiple_files       | OK      |        |
| list_directory__e8cf9eda.json             | list_directory            | OK      |        |
| list_directory__0c5f5c79.json             | list_directory            | OK      |        |
| list_directory__fe0dc6dc6.json            | list_directory            | OK      |        |
| list_directory__518341a7.json             | list_directory            | OK      |        |
| list_directory_with_sizes__e8cf9eda.json  | list_directory_with_sizes | OK      |        |
| list_directory_with_sizes__0c5f5c79.json  | list_directory_with_sizes | OK      |        |
| list_directory_with_sizes__fe0dc6dc6.json | list_directory_with_sizes | OK      |        |
| list_directory_with_sizes__518341a7.json  | list_directory_with_sizes | OK      |        |
| directory_tree__e8cf9eda.json             | directory_tree            | OK      |        |
| directory_tree__0c5f5c79.json             | directory_tree            | OK      |        |
| directory_tree__fe0dc6dc6.json            | directory_tree            | OK      |        |
| directory_tree__518341a7.json             | directory_tree            | OK      |        |
| search_files__580853a9.json               | search_files              | OK      |        |
| search_files__83e4c665.json               | search_files              | OK      |        |
| search_files__2831e0f1.json               | search_files              | OK      |        |
| search_files__ee546753.json               | search_files              | OK      |        |
| get_file_info__e8cf9eda.json              | get_file_info             | OK      |        |
| get_file_info__0c5f5c79.json              | get_file_info             | OK      |        |
| get_file_info__fe0dc6dc6.json             | get_file_info             | OK      |        |
| get_file_info__518341a7.json              | get_file_info             | OK      |        |
|                                           |                           |         |        |

| FIXTURE                                 | TOOL                     | VERDICT | DETAIL |
|-----------------------------------------|--------------------------|---------|--------|
| list_allowed_directories__44136fa3.json | list_allowed_directories | OK      |        |

## Leave-behind: your regression gate

The recorded fixtures (demo/fixtures–filesystem, suite fingerprint sha256:0764fd6d84d392dbff575042634e29e75bfa82344acc43b68759ae9166d7c478) are the behavioural contract of this server. Keep them in the repository and run the workflow below in CI — any breaking or value drift fails the build before it reaches your users.

```
# The recorded fixtures in demo/fixtures–filesystem are the behavioral contract: any drift fails this
name: mcp-proof regression gate
on:
  push:
  pull_request:
jobs:
  replay:
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@v4
      - uses: actions/setup-python@v5
        with:
          python-version: "3.11"
      - name: Install mcp-proof
        # not on PyPI yet – install from the repo that delivered this report:
        # pip install git+https://github.com/YuCPbit/mcp-proof
        run: pip install mcp-proof
      - name: Replay golden fixtures against the live server
        run: mcp-proof replay --fixtures demo/fixtures–filesystem -- npx -y @modelcontextprotocol/se
```

## Recommended next steps

- P1 LIFE-02 — Upgrade the server SDK to a release that speaks the 2026-07-28 spec revision.
- P1 SEC-04 — Add enum, pattern or maxLength to path/url/command-like string params.